Answers:

1. Initial review and backup :
    - Create a copy of data
    - Convert the range to an Excel Table so formulas and analysis expand automatically with new data.
2. Fixing inconsistent product name:
    - Use function TRIM() to remove extra spacing
    - Use function PROPER(),UPPER() or LOWER() to fix casing problem
    - Correct spelling by creating a reference (master) product list and using XLOOKUP() or VLOOKUP() to map incorrect names to standard names.
3. Fixing merged text values:
    - Use Text to Columns (delimiter-based) if the separator is the same in every row.
    - Use formulas such as LEFT(), MID(), RIGHT() when patterns are fixed.
    - Store them in separate clean column name as Region, Product and Code
4. Removing duplicates:
    - Remove duplicates by deleting repeated rows from unique sales id
    - If no unique id,use COUNTIFS() to flag true duplicates
    - Identify duplicates using Conditional Formatting and Highlight Duplicates then delete manualy if possible.
5. Removing text and extra symbols from numeric fields:
    - Use ISNUMBER() to validate numeric and flag non numeric rows in those columns.
    - Convert text to numbers using VALUE().
    - Remove symbols using SUBSTITUTE( ).
6. Data validation and summary statistics:
    - Apply Data Validation: Dropdown lists for Region and Product and Numeric limits for sales or quantity fields
    - Use AVERAGEIFS() and STDEV.S() with conditions to include only valid records.
7. Final analysis and reporting:
    - Prepare pivot table
    - Ensure all calculations reference structured table columns for dynamic updates.

To Prevent Future Errors:

- Data Validation: For Product, Region columns - Data > Data Validation > List - create dropdown.
- Excel Table: Convert entire data to Ctrl+T - Table named Sales_Data. Any formula uses structured references, no range errors.
- Protected Template: Lock formula columns, allow entry only in input columns.
- Standardize data entry templates and provide clear guidelines to all teams.
- Periodically review flagged invalid or duplicate records.

Ensuring accuracy with new data

● Because formulas, validations, and Pivot Tables are built on Tables and reference lists, new rows automatically inherit:
○ Cleaning logic

- Validation rules
- Correct aggregation
- Refresh Pivot Tables to update summary statistics without manual rework